

Note on Quantum Hall Effect

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Abstract

This is a note on QHE, based on David Tong's lecture notes.

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1 Preparations

First we'll give some preparations for the discussion of the QHE. We'll start with some basic concepts in CMT and QM. I won't give a detailed explanation, yet just list out some important results as a reference for later use. Most contents are explained in [1].

1.1 Band Theory and Conductivity

In a crystalline solid, the periodic ionic potential causes the allowed electron energies to split into **energy bands**, separated by **band gaps**. The key results are:

Theorem 1.1

Bloch's Theorem

The eigenstates of an electron in a periodic potential $V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})$ take the form

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{\mathbf{k}}(\mathbf{r}), \quad (1)$$

where $u_{\mathbf{k}}$ shares the lattice periodicity $u_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{\mathbf{k}}(\mathbf{r})$ and $\mathbf{k}$ lives in the first Brillouin zone.

Solving the Schrödinger equation with Bloch boundary conditions yields a discrete family of dispersion relations $E_n(\mathbf{k})$, labelled by the band index n . Each band accommodates $2N$ electrons (N = number of unit cells in the object, factor 2 from spin).

Here is a rough sketch of the band structure:

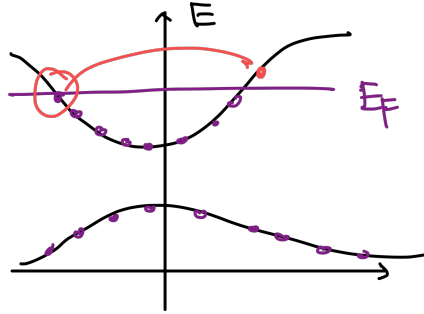


Figure 1: Band structure

Fermi energy and band filling. At $T = 0$, electrons fill all states up to the Fermi energy E_F .

- **Insulator / Semiconductor:** E_F lies inside a band gap; the valence band is completely full.
- **Metal:** E_F lies inside a band; there exists a Fermi surface.
- **Semimetal:** valence and conduction bands overlap slightly.

1.2 Landau Levels

1.2.1 Landau Gauge

Consider an electron moving in a 2D plane under a perpendicular magnetic field B . The Hamiltonian in Landau Gauge is:

$$H = \frac{1}{2m} \left(p_x^2 + (p_y + eBx)^2 \right) \quad (2)$$

Then the energy spectrum is quantized into **Landau levels**:

- **Wave Functions**

$$\psi_{n,k}(x, y) \sim e^{iky} H_n(x + kl_B^2) e^{-(x + kl_B^2)^2 / 2l_B^2} \quad (3)$$

with $H_n \exp\left(-\frac{x^2}{2}\right)$ the n -th Harmonic oscillator wave function and $l_B = \sqrt{\hbar/eB}$ the magnetic length. And $n \in \mathbb{N}$ is the Landau level index, $k \in \mathbb{Z}$ is the momentum along y direction.

- **Energy Spectrum** The energy of the n -th Landau level is:

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega_B \quad \omega_B = e \frac{B}{m} \quad (4)$$

And each level have a degeneracy of

$$N_B = eBA/2\pi\hbar \quad n = \frac{B}{\Phi_0} \quad (5)$$

which is the number of flux quanta through the system. These energy levels can be drawn in the following CMT style:

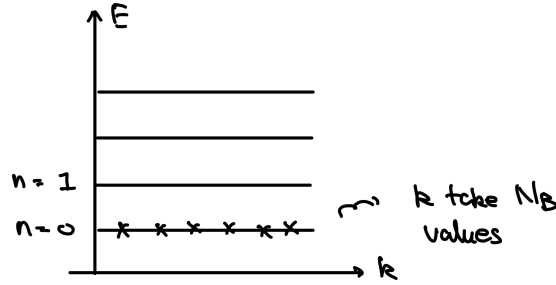


Figure 2: Landau levels

1.2.2 Landau Levels with Electric Field

If we add a uniform electric field E along x direction, then the Hamiltonian becomes:

$$H = \frac{1}{2m} \left(p_x^2 + (p_y + eBx)^2 \right) - eEx \quad (6)$$

Then we solve the Schrödinger equation and find the energy spectrum is still quantized into Landau levels, but with a shift:

- **Wave Functions**

$$\psi(x, y) = \psi_{n,k}(x - mE/eB^2, y) \quad (7)$$

- **Energy Spectrum**

$$E_{n,k} = \hbar \omega_B \left(n + \frac{1}{2} \right) + eE \left(kl_B^2 - \frac{eE}{m\omega_B^2} \right) + \frac{m}{2} \frac{E^2}{B^2} \quad (8)$$

We now see that different k states are not degenerate anymore, and the Landau levels are tilted in the energy-momentum space:

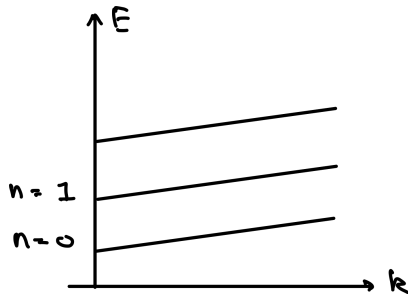


Figure 3: Landau levels with electric field

- **Group Velocity** If the wavefunction is a wave packet that is superposition of the Landau level states with different k , then the group velocity of the wave packet is:

$$v_y = \frac{1}{\hbar} \frac{\partial E_{n,k}}{\partial k} = \frac{E}{B} \quad (9)$$

it means that the wave packet will drift along y direction with a velocity.

1.2.3 Symmetry Gauge

YL: [To be completed, and important for IQHE]

1.3 Berry Phase

When a quantum system has a parameter dependent Hamiltonian $H(\lambda_i)$ and the parameters vary adiabatically along a closed loop C in the parameter space, the system's ground state wave function acquires a geometric phase known as the **Berry phase**.

1.3.1 Abelian Berry Phase

Consider the ground state energy is always normalized to 0, no matter which parameter we choose, then after a close loop of parameter variation, the wave function will acquire a phase factor:

$$|\psi(\lambda_i)\rangle \rightarrow e^{i\gamma} |\psi(\lambda_i)\rangle \quad (10)$$

where γ is the Berry phase, which can be expressed as an integral of the Berry connection A_i along the loop

- **Berry Connection and Curvature** The Berry connection is defined as:

$$A_i(\lambda) = -i \langle n | \frac{\partial}{\partial \lambda^i} | n \rangle \quad (11)$$

where $|n(\lambda_i)\rangle$ is a series of designated eigenstates of the Hamiltonian with fixed phase choices. And the Berry curvature is defined as:

$$F_{ij}(\lambda) = \partial_i A_j - \partial_j A_i \quad (12)$$

notice that i, j are indices of the parameters, not the spatial coordinates.

- **Gauge Transformation** if we change the choice of phase of the designated eigenstates by a gauge transformation $|n(\lambda_i)\rangle \rightarrow e^{i\alpha(\lambda)} |n(\lambda_i)\rangle$, then the Berry connection transforms as:

$$A_i(\lambda) \rightarrow A_i(\lambda) + \partial_i \alpha(\lambda) \quad (13)$$

which is similar to the gauge transformation of the electromagnetic vector potential. However, the Berry curvature is gauge invariant.

- **Berry Phase** can be expressed as the integral of the Berry connection along the loop:

$$i\gamma = -i \oint_C A_i(\lambda) d\lambda^i \quad (14)$$

or equivalently as the integral of the Berry curvature over the area enclosed by the loop:

$$i\gamma = -i \oint_S F_{ij}(\lambda) dS^{ij} \quad (15)$$

- **Chern Number** In a parameter space if we want the berry phase to be well-defined, then the integral of the Berry curvature over a closed surface must be quantized in units of 2π :

$$C = \frac{1}{2\pi} \oint_S F_{ij}(\lambda) dS^{ij} \in \mathbb{Z} \quad (16)$$

this integer C is called the Chern number, which is a topological invariant of the system.

1.3.2 Non-Abelian Berry Phase

When the system has degenerate ground states, the Berry phase becomes non-Abelian. We can see this from the fact that the designated eigenstates should span the degenerate subspace:

$$|n(\lambda_i)\rangle \rightarrow \{|n_a(\lambda_i)\rangle\}_a \quad (17)$$

Then we generalize the definition of Berry connection to a matrix-valued one.

- **Non-Abelian Berry Connection and Curvature** The non-Abelian Berry connection is defined as:

$$A_i(\lambda)_{ab} = -i \langle n_a | \frac{\partial}{\partial \lambda^i} | n_b \rangle \quad (18)$$

and the non-Abelian Berry curvature is defined as:

$$F_{ij}(\lambda) = \partial_i A_j - \partial_j A_i - i [A_i, A_j] \quad (19)$$

- **Gauge Transformation** if we change the choice of basis of the degenerate subspace by a gauge transformation $|n_a(\lambda_i)\rangle \rightarrow U_{ab}(\lambda) |n_b(\lambda_i)\rangle$, where $U(\lambda)$ is a unitary matrix, then the non-Abelian Berry connection transforms as:

$$A_i(\lambda) \rightarrow U(\lambda) A_i(\lambda) U^\dagger(\lambda) + i(\partial_i U(\lambda)) U^\dagger(\lambda) \quad (20)$$

Then we notice that the berry connection is a $U(N)$ gauge field, that take value of $u(N)$ Lie algebra. And the non-Abelian Berry curvature transforms as:

$$F_{ij}(\lambda) \rightarrow U(\lambda) F_{ij}(\lambda) U^\dagger(\lambda) \quad (21)$$

- **Berry Holonomy** The non-Abelian Berry phase (in fact its a matrix) can be expressed as a path-ordered exponential of the Berry connection along the loop:

$$U = P \exp \left(-i \oint_C A_i(\lambda) d\lambda^i \right) \quad (22)$$

2 Integer Quantum Hall Effect

Now we're prepared to discuss the integer quantum Hall effect (IQHE). The IQHE is a quantum phenomenon observed in two-dimensional electron systems subjected to low temperatures and strong magnetic fields.

2.1 Experimental Phenomenon

Here is a typical experimental setup for observing the IQHE:

Bibliography

- [1] D. Tong, “Lectures on the Quantum Hall Effect.” Accessed: Mar. 31, 2026. [Online]. Available: <http://arxiv.org/abs/1606.06687>